

MIKE HALL

Staff Software Engineer / Systems Architect

Chicago, IL | Email: mike@just3ws.com | Phone: (847) 877-3825 | Website: <https://www.just3ws.com> | LinkedIn: <https://www.linkedin.com/in/just3ws/> | GitHub: <https://github.com/just3ws>

PROFESSIONAL SUMMARY

Staff Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines hands-on software engineering with platform architecture, OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, and cloud infrastructure.

CORE SKILLS

Platform Architecture & Legacy Modernization, Cross-Domain Boundary Architecture & Governance, OpenTelemetry & Distributed Trace Context Propagation, High-Consequence Incident Escalation & Root Cause Analysis, Zero-Downtime Data Migrations & Schema Partitioning, Ruby on Rails, PostgreSQL, Sidekiq & Redis, AWS, Docker, Kubernetes & CI/CD Pipelines, Privacy-Conscious Local AI Orchestration & Tooling

EXPERIENCE

ASSOCIATE DIRECTOR, STAFF ENGINEER

OneMain Financial

Remote | January 2021 - February 2026

Led modernization of the Acquisition Lane, the digital front door for customer application and origination funnel workflows, delivered an instant disbursement workflow, applied Architecture Discovery to a silent application-completion defect, and aligned distributed observability with SRE operations.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.
- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

SENIOR BACKEND DEVELOPER

SK Holdings, Inc.

Chicago, IL | January 2019 - December 2020

Modernized search and asynchronous messaging pipelines under live production load, executing zero-downtime Rails framework upgrades and database relational indexing.

- Search Infrastructure Modernization: Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- Zero-Downtime Rails Upgrades: Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

SENIOR SOFTWARE DEVELOPER

ActiveCampaign

Chicago, IL | September 2018 - December 2018

Improved testability and performance inside a tightly coupled PHP/MySQL CRM backend, introducing structured query patterns and cache strategies, and hardening the Ember.js frontend test suite across core contact management workflows.

- Legacy Query Modernization: Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- Frontend Verification Gates: Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

SOFTWARE ENGINEER & TECHNICAL ONBOARDING LEAD (FRAUD & TAXONOMY SYSTEMS)

Groupon

Chicago, IL | July 2011 - May 2013

Implemented a centralized merchant taxonomy service in Java and MySQL, engineered analytical fraud detection in Vertica, and led global technical onboarding curricula during hyper-growth.

- Centralized Taxonomy Service: Implemented merchant taxonomy service in Java and MySQL, eliminating categorization drift across distributed product teams during hypergrowth.
- Analytical Fraud Detection: Built analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure for transaction anomaly scoring.
- Merchant Analytics Pipelines: Built high-throughput merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- Global Engineering Enablement: Redesigned technical onboarding curricula, standardizing engineering practices and shortening time-to-first-commit for 100+ global engineering hires.

ADDITIONAL EXPERIENCE

Senior Software Developer | BenchPrep (March 2017 - February 2018)

Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.

Senior Software Developer | ReachLocal (March 2015 - November 2016)

Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.

Open-Source Transition Lead | Coderwall (January 2014 - December 2014)

Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.

Principal Consultant | Tandem (August 2018 - August 2018)

Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.

EARLIER EXPERIENCE

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that inform my leadership today.

Revenue-critical commerce: Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.

Consulting and software craftsmanship: Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.

Enterprise and operational systems: Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.